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and Actuarial Science

MA981 DISSERTATION

BUILDING AN INTELLIGENT TRADING AGENT FOR THE NIGERIAN STOCK MARKET USING DEEP REINFORCEMENT LEARNING

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Abstract

This work investigates the application of Deep Reinforcement Learning (DRL) to algorithmic trading in the Nigerian Stock Exchange (NGX) through the use of the Proximal Policy Optimization (PPO) algorithm. A historical NGX stock data (2000-2025) was used to develop a trading environment with some of the most significant technical indicators such as Simple Moving Averages (SMA), Relative Strength Index (RSI), and Moving Average Convergence Divergence (MACD). The PPO agent was trained and evaluated against a buy and hold benchmark strategy.

The result shows how the PPO agent accumulated a total return of 1.4163, growing an initial capital of ₦10,000 to approximately ₦24,163, beating a buy and hold portfolio that grew to ₦11,480.07. While beating the benchmark considerably in returns, the Sharpe ratio of 0.1648 signifies excessive volatility and risk exposure. This implies that while PPO based trading rules can detect profitable trends in the Nigerian stock market, their risk adjusted performance must be enhanced.

The research concludes that such improvements as hyperparameter tuning, increased training timesteps, additional technical indicators, incorporation of macroeconomics, and risk management techniques are needed for attainment of improved robustness and real world applicability. The findings contribute to growing literature on DRL within the context of frontier markets and suggest that, with improved development, intelligent trading agents potentially hold a position to transform financial decision making in the Nigerian situation.

Keywords : Deep Reinforcement Learning, Proximal Policy Optimization, Nigerian Stock Exchange, Algorithmic Trading, Sharpe Ratio, Financial Markets

CHAPTER 1

Introduction

1.1 Background of study

In the last decades, financial markets have experienced a radical transformation with the development of algorithmic and data driven trading systems. What was once the domain of human traders who relied on intuition to make decisions has now become a universe governed by automation, quantitative models, and artificial intelligence (AI) [1]. One of the most promising developments in this field is the application of deep reinforcement learning (DRL) a machine learning paradigm that allows agents to learn optimal actions through interactions with their environment. Although developed economies such as the United States and China have seen dramatic development in intelligent trading systems, emerging economies such as Nigeria are underdeveloped. The Nigerian Stock Exchange (now known as the Nigerian Exchange Group or NGX) has unusual market characteristics of high inflation, interest rate volatility under the influence of policy, and structural inefficiencies. Nigerian trading practices remain manually or rule based with minimal use of intelligent trading agents. This provides an opportunity to explore how DRL can be utilized to automate and improve trading outcomes in such a volatile and data constrained market.

The Nigerian Stock Exchange (NSE) is an essential component of the nation's financial sector. It aids in the control of funds and propels economic development. Established in 1960, it is West Africa's oldest stock exchange. It has grown to be pivotal

to the region's economic development over the years through the provision of a central location for trading and the display of current stock prices reflecting Nigeria's general economy. The NSE has faced numerous challenges such as economic volatility, poor investor confidence, and narrow access to capital, which have slowed its expansion and general trading activity, Nonetheless, recent developments point to a dramatic change in technology enabled investment platforms, especially among young investors who prefer mobile trading apps and online brokerage platforms, This not only demonstrates an increasing desire for convenience but also a move towards socially responsible investing, where investors want their portfolios to reflect ethical and sustainable business practices. In order to enhance investment strategies and promote market advancement, individuals involved are increasingly considering the application of machine learning techniques to the stock market. Research has indicated that models such as Autoregressive Conditional Heteroskedasticity (ARCH) can be utilized to better comprehend market fluctuations, allowing investors to hedge against risks in periods of high market volatility. As the market evolves, new regulations aimed at enhancing transparency and safeguarding investors are likely to attract more individuals to participate, which will have a positive impact on the functioning of the market. All these developments collectively indicate an exciting period for the Nigerian Stock Exchange, characterized by growing utilization of technology and innovative investment approaches that may transform the way investors participate in the region.

1.1.1 The Capital Market as an Agent for Economic Growth in Nigeria

Capital market is an important sector of economic growth since it is an instrument of mobilization of long term finance of surplus sectors to the deficit sectors of an economy. It channels saving into productive investments and hence induces industrial growth, employment generation, and macroeconomic stability. In Nigeria, the history of capital markets, and specifically the evolution of the Nigerian Stock Exchange (NGX), is a reflection of the nation's economic experience from an earlier pre independence agricultural phase to a rapidly expanding oil based economy and, more recently, a technology innovation center. The establishment and subsequent growth of the NGX have played a fundamental role in providing a formal, regulated environment for companies to raise capital and for local and foreign investors to hold a stake in the

economic prosperity of the country. Besides the traditional equity market, the Nigerian economy offers alternative investment instruments, each with its own particular role in an ideally diversified portfolio.

1.1.2 The Genesis and the Early Years of the Nigerian Stock Exchange (1946-1977)

The formal history of Nigeria's capital market dates back to 1946 when the first government development loan stock was floated. However, it was the need for a structured and accessible market for the raising of capital for the public as well as private sectors that culminated in the creation of a formal stock exchange. The Lagos Stock Exchange (LSE) was established on September 15, 1960, a timely move made on the eve of the country's independence. Trading commenced on August 25, 1961, on a list of 19 securities, consisting of 3 equities and a mixture of government bonds and industrial loans (Nairametrics, 2020). The early days were marked by hand to hand, floor dealing and limited players in the market, with most trading within the premises of the Central Bank's building in Lagos.

The Capital Issues Committee was formed in 1962 to set rules for issuing securities, a precursor to the Securities and Exchange Commission. The government policies were a major driver of the early development of the market. The Nigerian Enterprises Promotion Decrees of 1972 and 1977, alternatively known as indigenization policies, were among the major factors that spurred the development of the exchange. The decrees mandated the transfer of foreign firms' majority share ownership to Nigerians, compelling numerous multinational corporations to float shares on the LSE (Nairametrics, 2020). This period saw the boom in listed companies, and the exchange was transformed from a weak, toddler market to a more robust platform. The LSE expanded beyond Lagos, and it was renamed the Nigerian Stock Exchange (NSE) on 2 December 1977 to reflect its national mandate. The NSE began opening up branches in principal commercial towns across the country, first of all in Kaduna in 1978 and Port Harcourt in 1980, later on in Kano, Ibadan, Onitsha, and Abuja.

1.1.3 From Deregulation to Digitalization: The Modern Era of the NGX (1986-2021)

The late 1980s witnessed the beginning of a new era for economic policy in Nigeria with the coming of the Structural Adjustment Programme (SAP). SAP, led by the World Bank and the International Monetary Fund, attempted to liberalize the Nigerian economy and wean it off state intervention. Under the capital market, SAP led to the removal of the indigenization decrees as well as privatization and commercialization of state owned enterprises. This opened the market up to the foreign investor and precipitated a wave of new listings, particularly from the banking and financial services sectors. The number of listed companies on the NSE grew from 42 in 1978 to 196 in 1999, an unmistakable sign of the impact of such reforms (Nairametrics, 2020).

Technological advancement became a priority for the NSE in the late 1990s. With a view to maximizing efficiency and transparency, the exchange moved from the manual trading platform to an Automated Trading System (ATS) on April 27, 1999. The computerized system supported real time trading and reduced transaction time dramatically. The new democratic dispensation of 1999 also raised investor confidence, attracting more capital inflows. The early 2000s saw the exchange enhance its technological development to a peak with the launch of the X-Gen trading platform in 2013. X-Gen was a new generation trading system designed to facilitate electronic trading for a broader group of investors, enhancing market access and liquidity. The pension reform was the other major development at the time. The Pension Reform Act 2004 created a new pension fund administration regime, which introduced long term and stable capital into the Nigerian capital market, as some of the funds were mandated to be utilized to invest in a variety of securities on the NGX (Nairametrics, 2020).

The most recent and groundbreaking watermark in the history of the exchange was its demutualization in 2021. The demutualization process converted the Nigerian Stock Exchange from a not for profit, member owned mutual organization to a public limited liability company, the Nigerian Exchange Group (NGX Group) Plc. This reorganization strategy formed three distinct and fully owned subsidiaries: Nigerian Exchange Limited (NGX), the trading exchange; NGX Regulation Limited (NGX RegCo), a standalone regulatory entity; and NGX Real Estate Limited (NGX RelCo). This demutualization

was aimed as a strategic step to release value, improve corporate governance, and make the exchange more competitive globally. Currently, NGX is a key member of the World Federation of Exchanges (WFE) and founding member of the African Securities Exchanges Association (ASEA), further solidifying its position as a key player in the African financial space.

1.1.4 Beyond Equities: A Broader View of Nigeria's Investment Climate

The Nigerian financial market extends far beyond the equity market, and there are numerous investment instruments to suit different risk appetites, investment horizons, and financial goals.

- Bonds and Fixed Income Securities: These are debt securities that represent a loan from an investor to a borrower, typically a government or a corporation. They are a cornerstone of any diversified portfolio, offering a steady source of income and comparative safety.
  - Government Bonds: They are issued by the Federal Government of Nigeria (FGN) and are the most secure investment in Nigeria since they are supported by the government's full faith and credit. They are made up of FGN Bonds and Treasury Bills, which are short term government securities.
  - Corporate Bonds: These are released by private companies and offer higher yields than government bonds to compensate for the higher credit risk. They are a significant source of long term capital for corporate expansion and are quoted on the NGX bond market.
  - Sukuk Bonds: A relatively new but fast evolving market in Nigeria, Sukuk bonds or Sharia compliant bonds are investment instruments. They represent an ownership interest in an asset rather than a debt and pay returns based on a share of profit generated by the asset.
- Real Estate: Real estate is a highly tangible and real form of investment for the average Nigerian. It is capable of providing both capital appreciation and steady rental income. Investment options include:

  - Direct Property Ownership: This involves buying property or land for commercial or residential purposes, or land banking.
  - Real Estate Investment Trusts (REITs): REITs are entities which invest, own, or provide financing for income producing real property. They provide a cheap way for investors to put money into the real estate industry without the trouble of direct management since they are stock exchange listed.

- Collective Investment Schemes: Such schemes pool funds from multiple investors to invest in a variety of diversified securities

  - Mutual Funds: Under the guidance of skilled fund managers, mutual funds offer instant diversification and are an ideal entry point for small investors. They are divided based on their investment objective, e.g., money market funds (short term, low risk), bond funds, and equity funds.
  - Exchange Traded Funds (ETFs): ETFs are mutual funds but trade on the stock exchange just like regular stocks. They allow for real time price discovery and greater liquidity than conventional mutual funds.

- Alternative Investments: The Nigerian investment landscape is also seeing growth in alternative assets, which are driven by technological innovation and portfolio diversification.

  - Agriculture: In the wake of agritech platforms, it is possible for investors to invest in agribusinesses under a crowdfunded model. Platforms allow people to fund certain agriculture projects, say growing crops or breeding livestock, in exchange for an entitlement to a proportion of returns.
  - Cryptocurrency: Despite regulatory issues, cryptocurrencies have gained widespread popularity as a high risk, high reward investment. Investors purchase and trade digital currencies like Bitcoin and Ethereum on various peer to peer platforms.
  - Fintech and Startups: Nigeria's growing tech sector offers the potential to invest in sound startups, sometimes through angel investing networks or venture capital expert funds.

The Nigerian financial market is a constantly evolving and dynamic place. The transformation of the Nigerian Stock Exchange, from a nascent, pre-independence entity into the demutualized, technology driven Nigerian Exchange Group, is a testament to Nigeria's economic strength and commitment to the development of a modern capital market. The turning points beginning with the indigenization initiatives of the 1970s and liberalization of the 1980s, to the tech boom of the 1990s and demutualization in more recent times have, collectively, developed the NGX into a key force for economic growth. More significantly, the diversification of investment tools beyond the traditional equity market reflects the maturity and sophistication of the Nigerian financial environment. From the security of government bonds and the stability of property to the high growth potential of mutual funds and alternative assets, the market provides a solid platform for raising funds as well as wealth creation. A wise trading agent for this market, therefore, ought not only be well informed about the processes of trading equities but also about a deeper context for these diverse options for investment and the long term historical forces that have produced today's market form.

1.2 Problem Statement

Notwithstanding increasing international attention in AI driven trading strategies, Nigeria mainly depends on rule based or manual methods. There exists a significant research gap in the application of DRL particularly policy-gradient techniques such as Proximal Policy Optimization (PPO) for the optimization of trading in frontier markets such as the NGX, which is volatile and has low liquidity [2]. In addition, there is a lack of research in the academic and industry literature investigating the application of deep reinforcement learning in African financial markets. An agent capable of learning from historical stock data, identifying worthwhile trading opportunities, and adapting to market trends would be a valuable novel concept for both the local finance community as well as broader research communities.

1.3 Research aim and objectives

This study aims to create and test a trading agent that uses deep reinforcement learning. This agent will learn to decide whether to Buy, Hold, or Sell based only on stock price

data from the Nigerian Exchange (NGX). The goals are as follows.

- Develop and implement a PPO based trading agent with NGX past price data.
- Create an environment and reward function
- Compare performance with conventional strategies such as Buy and Hold or Moving Average.
- Examine strengths and weaknesses, particularly in an emerging market environment.

1.4 Research Questions

This research seeks to address these critical questions.

- Can a PPO based deep reinforcement learning agent learn profitable trading strategies using Nigerian stock price data?
- What are the issues and limitations of applying DRL to a new market such as Nigeria?

1.5 Significance of the Study

This study is significant both for practical application and for research purposes.

- For brokers and fin-tech businesses, the creation of a DRL based trading agent can be a base model for smart investing platforms and robo-advisors that are tailored to Nigerian market conditions.
- For traders and investors, these agents can simplify complex decisions and minimize emotional bias that can influence trading.
- For academia, the research adds to the scarce existing research on AI applications in African financial markets, laying the groundwork for further research on emerging economies.

1.6 Scope and Limitations

The project only considers NGX stock data between the years 2000 and 2025. To make it manageable, it considers a limited set of stocks, and only price-based technical indicators are part of the model. Macroeconomic variables (such as inflation and bond yields) are excluded at this stage and are recommended for future study. Potential problems include gaps in the simulation environment, data quality issues, and fitting too closely to historical price trends.

CHAPTER 2

Literature Review

2.1 Overview of Algorithmic Trading and DRL in Finance

Algorithmic trading has moved on from rule based systems e.g., moving average crossovers and RSI levels to more reactive, adaptive frameworks based on machine learning and, more recently, deep reinforcement learning (DRL), DRL brings together reinforcement learning and deep neural networks in a unique way, allowing agents to make sequential decisions in complicated, high dimensional environments. Notably, DRL applications have gained more use in sectors like gaming and robotics, and there is an increase in exploring DRL in financial settings (Pricope, 2021).

Traditional quantitative models likewise struggle to keep pace with the dynamic, highly volatile changes characteristic of stock markets. This has encouraged studies on more adaptive and robust methodologies, some of which include machine learning, and Deep Reinforcement Learning (DRL) in general, which have turned out to be particularly promising. This review addresses the progression from traditional trading frameworks to leveraging advanced DRL algorithms, specifically referencing Proximal Policy Optimization (PPO) as a state of the art peak performance algorithm for designing an independent trading agent.

2.2 From Machine Learning to Reinforcement Learning in Finance

- Machine learning in finance is nothing new. Classic supervised models such as Support Vector Machines (SVMs) and Long Short-Term Memory (LSTM) networks have been used extensively on tasks such as price forecasting and sentiment analysis. However, these are handicapped by their reliance on fixed labeled data and inability to make sequential forecasts in the dynamic world. They can predict a direction but not formulate an entire trading policy that decides when and how much to trade in the long term (Deng et al., 2017).

- Reinforcement Learning (RL) is a better paradigm. In the RL paradigm, an agent (the trading algorithm) learns to make the optimal decision by trial and error based on feedback from an environment (the stock market). The agent observes the state (market prices, trading volume, technical indicators), takes an action (buy, sell, or hold), and receives a reward (profit or loss). The goal of the agent is to learn a policy which is a collection of rules for choosing actions that maximizes its cumulative reward in the long term. By this dynamic, interactive trial and error learning process, the agent can develop dynamic strategies capable of adapting to diverse market situations without the need for implicitly defined rules or tagged data (Sutton and Barto, 2018).

2.2.1 Deep Reinforcement Learning: A New Paradigm for Trading

Traditional RL methods have a problem with high dimensional state space, such as the vast amount of data generated in the financial markets. The breakthrough came when deep neural networks were invented, which led to the field of Deep Reinforcement Learning (DRL). DRL uses deep neural networks to approximate the value function or policy and enables the agent to learn directly from raw, high level market data (Mnih et al., 2015). This end to end learning allows the agent to find and exploit subtle, non-linear patterns that are not accessible to humans or simpler models.

The resilience of DRL in finance is the ability to solve the sequential decision making issue of dealing in an uncertain world. Théate and Ernst (2020) demonstrated that a

DRL agent was able to learn profitable actions in a simulated stock trading environment, successfully balancing risk and transaction costs. The DRL framework allows for the optimization of a clear objective function, e.g., profit maximization or a risk adjusted return metric such as the Sharpe ratio, and hence is particularly appropriate for quantitative trading.

2.2.2 Proximal Policy Optimization (PPO)

Within the family of DRL algorithms, policy gradient methods are highly applicable to trading since they can learn stochastic policies in continuous action spaces. They are, however, often plagued by destructively large policy updates, which result in unstable training. Proximal Policy Optimization (PPO) by Schulman et al. (2017) addresses this critical issue of instability.

PPO is an on-policy, policy gradient approach that improves training stability by constraining the size of policy updates at each step. It accomplishes this through a "clipped surrogate objective function," which punishes the new policy for deviating too far from the old policy. This clipping leads to more stable and monotonic performance improvement and makes it extremely well suited for noisy and volatile environments like stock markets. Thanks to its reliability, sample efficiency, and ease of implementation, PPO has become a go to option for numerous DRL applications.

2.3 DRL in Algorithmic Trading: Evidence and Strategies

- Ensemble Strategies : Yang et al. (2020) suggested a DRL ensemble that unites PPO, A2C, and DDPG for stock trading optimization. The approach obtained better risk adjusted returns (estimated through Sharpe ratio), surpassing individual agents and conventional baselines. Similarly, comparative studies typically find that PPO demonstrates a better performance stability trade-off compared to other DRL algorithms in cases of portfolio management and trading execution (Lucarelli and Borrotti, 2019).

- Architectures and Reward Design : Théate and Ernst (2020) proposed TDQN (Trading Deep Q-Network), optimizing directly for the Sharpe ratio instead of raw returns. This reward based approach proved to be robust in various market config-

urations (Théate & Ernst, 2020). Zou et al. (2022) proposed a cascaded LSTM-PPO (CLSTM-PPO) framework. By coupling sequence modeling through LSTM with PPO's policy optimization, the model achieved considerable improvement in cumulative returns and risk adjusted performance particularly in emerging markets such as China. An effectively designed reward function is consistent with the actual objective of the trading agent in reality (i.e., risk adjusted return maximization, not raw profit). Several papers suggest including financial measures like the Sharpe ratio, Sortino ratio, or risk penalties directly within the reward signal to encourage less volatile, more stable strategies (S. Lawton and Asad Jeewa 2024), (I. Varshini Devi et al. 2023). This can generate more stable out performance relative to profit and loss based reward structures. Now using PPO, PPO is an actor-critic algorithm that aims for a balance between policy stability and convergence speed. It is well known for learning robust trading policies in noisy and non-stationary market environments (Hima Keerthi Sagiraju and Shashi Mogalla 2021). PPO's clipped objective functions and sample efficient updates are especially well suited for financial time series (M. Vishal, Yatin Satija, and B. Babu 2021).

- DRL in Emerging Markets: Nigeria : Market specific research for Nigerian or African markets is still limited. One study, however, used DDPG based methodologies in a market-neutral setup with CNN and RNN models and showed that reward formulations that include the Sharpe ratio provided around 11 percent more superior results than return-only rewards (Olabode, 2022).

- Reviews: Pricope's survey (2021) is still one of the most extensive summaries of DRL in finance, with a focus on methodological constraints and the disconnect between academic proof-of-concepts and production deployments. A more recent meta analysis by Bai et al. (2024) identifies the main challenges across RL in finance e.g., financial market modeling as MDPs, explainability, robustness, and realistic simulation environments and proposes future research directions such as contextual RL and multi-agent systems.

2.4 Summary of past literature and Implications

Deep reinforcement learning has shown success in controlled settings, and advanced and ensemble architectures gives performance boosts, it is also important to note that Optimizing for risk adjusted returns example; Sharpe ratio, improves agent robustness. Emerging markets like the Nigerian market, has very few studies targeted at it. This makes this study very relevant and unique.

The literature overwhelmingly supports the use of Deep Reinforcement Learning for building intelligent trading agents. DRL algorithms have consistently been shown to learn high fitness profitable policies directly from the market data, outperforming standard baselines. Proximal Policy Optimization is among the top DRL algorithms by linear stability and robustness, so it's a leading choice for navigating the underlying volatility of financial markets.

There does exist, however, a gap in the current body of literature. Most of the studies utilizing DRL and PPO in algorithmic trading are for large, liquid markets such as the NYSE, NASDAQ, and FOREX. There is a dearth of studies examining whether these high tech methods are applicable and effective in emerging markets such as the Nigerian stock market. This dissertation aims to bridge this gap by formulating, designing, and experimenting with a PPO based trading agent particularly tailored for the unique character and dynamics of the Nigerian stock market, thereby contributing something novel towards the use of DRL in less researched financial systems.

CHAPTER 3

Methodology

3.1 Research Design

This study employs a quantitative and experimental method, focusing on the simulation of a stock trading agent specifically designed for the Nigerian Stock Exchange (NGX) using Deep Reinforcement Learning (DRL). The agent is trained in an environment created to mimic real world trading conditions. The main algorithm used is Proximal Policy Optimization (PPO), a state of the art DRL technique that has been acclaimed for its solidity and efficiency in solving continuous as well as discrete decision making problems. This study tends to be unique as there has not been alot of studies using reinforcememnt learning or deep reinforcement learning on financial datasets within emerging markets like Nigeria.

The methodological workflow consists of several stages;

- Data Preparation and Loading: At this stage the data set used was loaded and prepared for work
- Data Preprocessing and cleaning: At this stage the dataset is preprocessed and cleaned, setting date as the index and handling missing values
- Feature Engineering and Calculating Technical indicators: Technical indicators like Simple Moving Average (SMA), Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD) which are used by traders and financial

institutions to analyze market trends and predict price movements are calculated and added to the datasets also the data is scaled to have numerical variables fall within a certain range.

- Data splitting: We then split the data into training (70 percent), validation (15 percent), and testing (15 percent) subsets to avoid overfitting and assess generalization.

- Creating the custom trading environment

- Initialize and Train the agent: The stable baseline3 library is used to initialize the agent and the agent is the trained on the custom environment

- Evaluating the agent: The Trained agent is evaluated on the testing data and the cummulative return and sharpe ratio is calculated.

3.1.1 The Concept of Machine Learning, Reinforcement Learning and Deep Reinforcement Learning

Machine learning (ML) is a branch of artificial intelligence that focuses on designing systems that learn from experience. Machine learning systems can be categorized broadly into three main categories, distinguished on the basis of the type of data and the objective of learning.

- Supervised Learning: It is the most widely used, where the algorithm learns from a set of labeled examples. The example consists of input features and the correct output (a "label" or "target") for those features. The aim is to acquire a mapping function that can provide the right output for new, unseen inputs. In finance, it could be forecasting whether the stock price goes up or down (classification) using historical technical indicators or data.

- Unsupervised Learning: In this case, the algorithm is shown data with no clear labels. Its task is to find hidden patterns, organization, or groupings within the data. Common tasks are clustering (e.g., grouping stocks by comparable volatility patterns) or dimensionality reduction (e.g., compressing numerous market indicators into a lower number of representative factors).

- Reinforcement Learning (RL): Unlike the other two, RL is concerned with learning to make a sequence of good decisions. An agent learns through interacting directly with an environment that can change over time and receiving feedback in the form of a reward or penalty for an action taken. It is not concerned with finding patterns or predicting labels but learning a behavioural strategy, referred to as a policy, to maximise the overall reward over time.

3.1.2 Limitations of Supervised and Unsupervised Learning for Algorithmic Trading

Though better for some analytical purposes, supervised and unsupervised learning are both flawed in their use to address the inherent issue with algorithmic trading.

The worst flaw of supervised learning is that it treats trading as a static prediction problem rather than a dynamic sequential decision problem. A model can accurately forecast the price of a stock will likely increase, but this alone cannot be used to create a profitable strategy. It fails to respond to important follow-up questions: How much to buy? When to sell in order to gain the profit? How does the transaction cost influence the decision? Each of these trading decisions affects the agent's future situation (e.g., money available) and future opportunities, a dependency which supervised models are not well suited to cope with.

Unsupervised learning, then, is descriptive, rather than prescriptive. It will reveal informative facts, such as the fact that a group of technology stocks cluster together. But it won't produce an actionable trading strategy. It can inform a strategy but can't act on one itself by making buy, sell, or hold suggestions. Reinforcement Learning, by its very nature, is able to overcome these limitations. It does away with the difficulty of sequential decision making inherent in trading by learning a unified policy from a mapping of market states to concrete actions to maximize a long term financial return.

3.1.3 From Reinforcement Learning (RL) to Deep Reinforcement Learning (DRL)

The basic RL model is marked by interaction between the agent and the environment. In this study:

- The Agent is the trading algorithm.
- The Environment is the Nigerian stock market.
- The State S_t is a description of the state of the market at the time t, and it may be described using data such as historical prices, volumes, and technical indicators.
- The Action A_t is the chosen action of the agent, i.e., purchasing, selling, or holding an asset.
- The Reward R_{t+1} is is the numerical reward obtained following the performance of an action, typically the actual profit or loss.

The goal of the agent is to discover an optimal policy π^* that maximizes the expected discounted cumulative reward, or return G_t defined as:

$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$

Where $\gamma \in [0, 1]$ is a discount factor that prefers immediate rewards to future ones. All of the most common traditional RL approaches, such as Q-learning, usually utilize an array (a Q-table) to store the expected return for every possible state-action pair. This can only be done for those problems that have a small, discrete set of states. A financial market's state space is, in reality, infinite and continuous. The number of feasible combinations of prices, volumes, and indicators is so vast that it is impossible to represent with a lookup table. This is known as the "curse of dimensionality."

Deep Reinforcement Learning (DRL) solves this problem through the use of deep neural networks within the RL framework. Instead of a Q-table, a DRL agent uses a deep neural network as an approximator of the policy or value function. This allows the agent to:

- Handle High-Dimensional State Spaces: Neural networks can directly handle raw market data without extensive feature engineering.
- Generalize from Experience: The network is capable of making intelligent decisions in situations in the market it has never before encountered by generalizing from similar past conditions.
- Learn Complex, Non-Linear Patterns: Neural networks are capable of modeling the highly complex and non-linear behavior observed in financial markets.

3.2 Trading Data

The data set used for this work was gotten from kaggle, it contains stock historical prices from 2000 up until 2025, containing over 13 sectors and 145 companies, with dialy dates, prices (open,high,low), percentage change,volume multiplier,volume,subsectors of each company, market classification, shares outstanding and market cap. The data contains over 307,582 daily stock prices across the different sectors and companies.

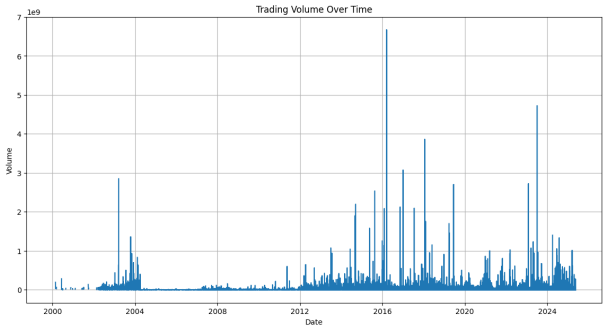


Figure 3.1: Trading Volume over Time

The figure above shows the trading volume over a time frame in the Nigerian stock market, this indicates that 2016 has the most trading activities, this shows that trading activities has fluctuated significantly over the years which is some what normal for stocks.

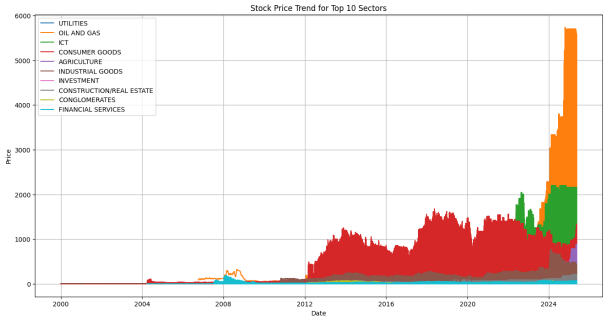


Figure 3.2: Stock Price Trend for Top 10 Sectors

The top 10 sectors by average dialy returns are Utilities, oil and gas,ICT, consumer goods, agriculture, industrial goods, investment, construction/real estate, conglomerate, financial services. The Figure 3.2 shows the stock price trends for the top 10 sectors over time.

Little activity is observed in the first decade. The total cost for all sectors is minimal, generally below 500. There is a minimal increase in activity around 2008, which could be indicative of market volatility during the Global Financial Crisis, but the overall impact of this scale is minimal.

From about 2012, there is a strong and persistent rise in the total price. This rise is primarily fueled by the CONSUMER GOODS sector (red). It grows quickly, becoming the overwhelming part of the total price. The other sectors constitute a fairly thin, stable foundation at the base during this period.

There is a trough and volatility around 2020 which is extremely likely to be the economic impact of the COVID-19 pandemic. Following this trough, the market returns, but the mix begins to change. Consumer Goods remains substantial, but other sectors, particularly ICT (green), begin to increase more dramatically.

This is the most dramatic feature of the chart. During the last year or so, the total price takes an explosive, near vertical flight, taking off from slightly over 2000 to nearly 6000. This rise is dominated by two sectors:

- OIL AND GAS (orange): This sector, which was negligible for the last 24 years, suddenly becomes the single largest contributor, representing the lion's share of the price increase.
- ICT (green): This sector also sees explosive growth over the same period, forming the second-largest component of the boom.

The previously dominant Consumer Goods sector, although still present, is completely overshadowed by this new boom in technology and energy.

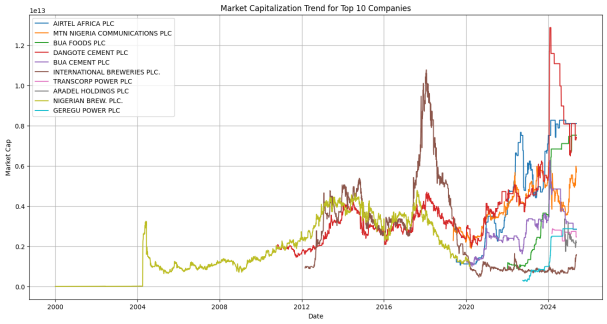


Figure 3.3: Market Capitalization Trend for Top Ten Companies

The top 10 companies by average market cap, in the Nigerian stock market are:

AIRTEL AFRICA PLC, MTN NIGERIA COMMUNICATIONS PLC, BUA FOODS PLC, DANGOTE CEMENT PLC, BUA CEMENT PLC, INTERNATIONAL BREWERIES PLC, TRANSCORP POWER PLC, ARADEL HOLDINGS PLC, NIGERIAN BREW. PLC, GEREGU POWER PLC. The figure 3.3 shows the market capitalization trend for this top 10 companies across the years. This chart presents a close up of the market capitalization path of ten stand-alone, high cap companies over 25 years from 2000 to mid-2025. It plots the stand alone market cap of each company as a single line, thus enabling direct comparison of their value and performance over time.

In the first decade, the graph shows very little information, with only one company, ARADEL HOLDINGS PLC (in olive), that is easily visible from around 2004. This would suggest either that the other companies had not been listed, were not yet large enough to be accounted for, or historical information is thin. From 2012 or so, more firms show up on the chart, and the market is highly dynamic. There is strong rivalry for the leadership position, mainly between DANGOTE CEMENT PLC (red) and MTN NIGERIA COMMUNICATIONS PLC (orange), with quite a number of other firms experiencing high but smaller peaks. There were several steep rallies and deep declines in this phase, reflecting a volatile market.

The most turbulent activity occurs from 2021 and onwards. DANGOTE CEMENT experienced a gigantic, steep peak, its highest point early in 2024, before experiencing a rapid decline. AIRTEL AFRICA PLC (blue) and MTN NIGERIA (orange) saw their market caps surge, vying for the number one spot with each other several times. BUA FOODS PLC (green) and BUA CEMENT PLC (purple) entered the market or had their valuations surge, becoming instant rivals and adding to the volatility. The most recent figures show a frenzied scramble for leadership by Airtel, MTN, Dangote Cement, and BUA Foods, all of whom have failed to keep up and maintain a clear lead.

3.3 Technical Indicators

SMA (Simple Moving Average), RSI (Relative Strength Index), and MACD (Moving Average Convergence Divergence) are all technical indicators that traders utilize to examine market trends and forecast price movement. SMA gives a general trend by taking the average of previous prices. RSI calculates the velocity and change of price

movements to determine overbought or oversold levels. MACD indicates the correlation between two moving averages to disclose trend direction and strength.

- Simple Moving Average (SMA): The SMA is a simple technical indicator that computes the average closing price of a security over a given time frame. It is a lagging indicator since it relies solely on past data, smoothing out short-term price volatility to reveal the general trend. it Identifies trends, A rising SMA indicates an uptrend, and a falling SMA indicates a downtrend. The SMA also Finds support and resistance, During an uptrend the SMA can serve as a dynamic support level at which the price of a stock tends to bounce higher after a small dip. The opposite occurs in a downtrend. Lastly, it helps generates buy/sell signals, Traders look for crossovers, like a "golden cross," when a shorter-term SMA crosses over a longer term SMA, to indicate a possible uptrend. For example; A 20-day SMA is the average closing price for the most recent 20 days. On each day, the oldest observation is removed and the latest one is included, so the average "moves" with the price.

- Relative Strength Index (RSI): The RSI is a momentum indicator that quantifies the speed and magnitude of price movement. It is represented as a number between 0 and 100 and is mostly utilized to detect overbought and oversold positions. It identifies overbought conditions, If the RSI is greater than 70, then the stocks is overbought and may be in line for a price correction. It also Identifies oversold conditions: If the RSI falls below 30, the stocks is oversold and may result in a price increase. It can also Spots divergence, Divergence happens when a stock's price goes in the opposite direction of the RSI, and it might indicate a potential reversal of the trend.

- Moving Average Convergence Divergence (MACD): The MACD is a trend directional momentum indicator that shows the relationship between two moving averages (EMAs) of a stock's price. It consists of three parts: MACD line, signal line, and histogram. It Shows momentum and trend reversals, which means it is used for identifying reversals in the direction or size of a price trend. It also generates buy/sell signals, When the MACD line crosses above the signal line, it gives a buy signal. When it crosses below, it generates a sell signal.

3.4 Trading Environment Design

The environment for trading is designed using the OpenAI Gymnasium interface. It allows the PPO agent to interact with market data within an organized cycle of observations, actions, and rewards.

- Observation Space: A state is a flat array of stock properties (Price, Volume, Change percentage) of a sliding window of 14 time steps. This allows the agent to see short-term market trends

- Action Space: he agent may take one of three discrete actions at each step:

  - 0 = Hold (do not transact)
  - 1 = Buy (use current balance to buy shares)
  - 2 = Sell (sell all owned shares)

- Reward Function: Defined as the percentage change in portfolio net worth:

  $r_t = \frac{W_t - W_{t-1}}{W_{t-1}}$

  where W_t is the agent's net worth at time t, This reward encourages strategies that maximize long-term wealth accumulation and discourages drawdowns.

- Termination: Episodes are terminated when the dataset is exhausted or when the agent reaches the terminal timestep. This environment simulates the financial decision making process of a trader, enabling the agent to learn profitable policies via interaction with historical NGX data.

- Market Frictions: Transaction Costs and Slippage To make the world of trading as realistic as possible, both transaction costs and slippage were introduced. Slippage works by modelling the divergence between intended trade prices and actual prices at which trades are made by incrementing buy prices and decrementing sell prices by a minimal amount. Transaction costs were placed as a percentage charge on all transactions. Both of these constraints penalise over trading, dampen unrealistically high profits, and compel the agent to use more efficient risk aware techniques.

3.5 Proximal Policy Optimization (PPO) Algorithm

3.5.1 Why PPO?

PPO is a cutting-edge policy gradient algorithm that searches for a balance between performance improvement and stability in training. In contrast to value-based methods such as DQN, which learn Q-values, PPO directly optimizes the policy (action probability distribution given states). This is particularly useful for stock trading because, The environment is stochastic and sequential, actions have long term consequences, also stable updates are more important not to overfit noisy financial signals.

3.5.2 PPO Objective Function

PPO generalizes vanilla policy gradients by possessing a clipped surrogate objective.

The typical policy gradient update maximizes:

$L_{PG}(\theta) = E_t[\log \pi_\theta(a_t|s_t)A_t]$

where $\pi_\theta(a_t|s_t)$ is policy (probability of taking action a_t in state s_t

A_t is advantage function (how good an action is relative to the average).

Large updates can, however, destabilize the training. PPO remedies this by establishing the probability ratio:

$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$

This ratio is between the new and old policies. PPO then maximizes:

$L_{CLIP}(\theta) = E_t[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)]$

where ϵ is a hyperparameter (typically 0.1–0.2).

- if r_t is within the range $1-\epsilon, 1+\epsilon$ then PPO behaves like normal or standard policy gradient.

- if r_t is beyond this range, the objective is clipped so that very big updates aren't done

This ensures the policy improves reliably without diverging, which is an important property when dealing with volatile stock data

3.5.3 Training Dynamics in the Trading Environment

- Interaction: PPO observes the window of stocks (state), samples an action (Buy/Hold/Sell), and receives a reward.
- Experience Accumulation: Transitions (state, action, reward, next state) are stored.
- Policy Update: After some steps, PPO updates its policy by maximizing the clipped surrogate objective carefully, making sure the updates are controlled.
- Repeat: The agent repeats until convergence, learning incrementally when to invest in rising markets, sell to harvest gains, and hold through noise.

3.5.4 Implementation Steps

- Environment Construction: Implemented in Python with gymnasium, creating observation, action, and reward functions.
- PPO Agent Initialization: Implemented using Stable-Baselines3 and MlpPolicy as the neural network architecture.
- Training: Trained on training data for more than 50,000 timesteps.
- Model Saving: Trained policy saved to evaluate later.

3.5.5 Expected Results

At the end of the training;

- PPO learns a policy network that projects past market windows to best trading actions.
- The agent trained will do better than random policies and even rival baseline policies like Buy-and-Hold
- Evaluation on the test set will reveal generalization capability, measured by means of cumulative returns and Sharpe ratio

CHAPTER 4

Analysis and Results

This chapter presents the result of the experiments carried out in building and testing the intelligent trading agent for the Nigerian Stock Exchange (NSE) using Proximal Policy Optimization (PPO). The explanation includes the preprocessing of raw stock market data, training the PPO-based trading agent in a purpose-built environment, and its testing using financial metrics such as cumulative return and Sharpe ratio. The objective is to assess whether a PPO based reinforcement learning agent can generate profitable trading signals using historical Nigerian stock market data and outperform naive baseline strategies such as buy and hold.

4.1 Data Preprocessing and Feature Engineering

The information utilized in this case is the NGX Stocks (2000-2025), which has historical stock price and volume information. For more informative learning of the PPO agent, corresponding technical indicators were computed, which are feature inputs to the trading environment:

- Simple Moving Averages (SMA): 20 day (SMA-20) and 50 day (SMA-50) were computed to detect short-term and long term trends of the prices.

- Relative Strength Index (RSI-14): Momentum indicator suggesting overbought or oversold market conditions.

- Moving Average Convergence Divergence (MACD): The 12 day exponential moving average minus the 26-day exponential moving average.
- MACD Signal Line: A 9 period exponential moving average of the MACD to signal changes in momentum

For enabling feature comparability, Min-Max normalization was employed, scaling all numerical features to a [0,1] interval. Missing values were treated with forward fill and dropping rows from the beginning where rolling averages were not calculable.

The preprocessing stage provided the agent with noise filtered, information dense, and normalized data for optimal learning.

4.2 Custom Environment

The environment is a simulation of a stock trading process using historical data from the Nigerian Stock Exchange.

- State (Observation): Each state is a flat vector of numeric features (like Price, Volume, Change in percentage, some technical indicator like Simple Moving Average (SMA), Relative Strength Index(RSI), Moving Average Convergence Divergence (MACD) over a window of size 14.
  - This captures short-term historical trends, which are useful for the agent to make good decisions.
  - Example: at step t, the agent sees prices and volumes for $t - 13$ to t.
- Action Space: Is discrete with three actions:
  - 0 = Hold (do nothing)
  - 1 = Buy (purchase as many shares as possible with available balance)
  - 2 = Sell (sell all shares owned)

  The agent goes all in when selling and the same happens when buying, except it decides to hold.
- Reward Function: Reward is percentage change in net worth between steps

  - If agent does increase the portfolio value = positive reward.
  - If portfolio value decreases = negative reward
  - This makes the PPO agent to learn policies that always make money in the long run.

- Termination: The episode ends when the dataset is depleted (last date of historical stock price data).

4.2.1 How PPO Learns in This Environment

PPO (Proximal Policy Optimization) acts with the Trading Environment as follows:

- Agent starts training:
  - At each step, PPO observes the current state (prices/indicators window).
- Policy (Actor) selects an action:
  - The agent computes probabilities of buying, holding, or selling
  - An action is sampled from this distribution.
- Environment applies the action:
  - If action = Buy then balance decreases, shares increase.
  - If action = Sell then balance increases, shares are set to 0.
  - If action = Hold then no balance or shares update.
- Environment provides reward and new state:
  - PPO receives reward (portfolio net worth changes).
  - PPO sees next observation window (new stock information).
- Policy update (the essence of PPO):
  - PPO then stores transitions (state, action, reward, next state) in the memory.
  - After collecting enough steps, PPO updates policy with clipped objective optimization: This helps not to perform large, destabilizing policy updates and also Offers stable learning in noisy stock data.

- Iterative training: This is done repeatedly across many time steps, Over time PPO learns the action sequences maximizing portfolio returns.

4.2.2 The Outcome of Learning

After training on the training data set:

- Trained Policy is Saved: The model is stored as ppo training agent (as a zip file). The file contains the trained neural network weights showing the learned policy.
- Desired Agent Behavior: The agent is expected to learn the following behaviors such as when to buy after rising momentum, when to sell to lock returns, and when to hold during noise. Since the reward is tied to portfolio net worth, the agent will have the motivation to aim for long term wealth gain versus immediate profit.
- Keeps portfolio variables such as:
  - Present balance
  - Stocks owned
  - Net worth
  - All-time high net worth

The Environment issues reward signals on percentage changes in net worth of portfolio from step to step. This structure ensures that the goal of the agent ought to be that of optimizing total portfolio value, rather than short-term price predictions.

4.3 Training the PPO Agent

The PPO agent was created using the Stable Baselines3 implementation with the default hyperparameters:

- **Discount factor (γ):** 0.99
- **Learning rate:** 0.0003
- **Clip range (ϵ):** 0.2

- **GAE λ:** 0.95
- **n_steps (rollout size):** 2048
- **Batch size:** 64

The discount factor determines how much the agent wants future rewards compared to immediate rewards. It's a value from 0 to 1. A value of 0.99 would make the agent extremely patient. It would strongly prefer planning for the future, so it is suitable for long, complex tasks with the goal very far in the future.

The clip range prevents the agent from changing its strategy (policy) too much at one update. It sets a "trust region" and says, "Don't make the new policy more than 20 percent different from the old one." PPO calculates the ratio of the probability of an action under the new policy to that under the old one. It then "clips" this ratio to limit it within the interval $[1 - \varepsilon, 1 + \varepsilon]$, A clip range of 0.2 guarantees that learning is very stable. It avoids the phenomenon of a lucky or unlucky experience leading to an enormous, catastrophic update of the policy of the agent. This incremental, step by step improvement is the reason why PPO is so reliable.

Learning rate determines the amount by which the agent's neural network weights are updated in each training update. A learning rate of 0.0003 is a relatively small and conservative learning rate. It's chosen as a conservative default to induce stable and uniform learning on a very wide set of tasks without drifting away from taking a couple of bad updates and sending the entire training process astray.

Generalized Advantage Estimation (GAE) is an algorithm for estimating how much better the specific action was compared to the average action at a specific state. The parameter sums different estimates to achieve an optimal trade off. A GAE of 0.95 results in a more refined blend that utilizes the whole sequence of rewards significantly but discounts them to some extent to reduce variance. This provides a good and stable estimate of an action's "advantage," which is crucial for effective learning.

The rollout size is the number of steps the agent will take in the environment to collect data before stopping to train its neural network. This collected data is what is called a "rollout" or a "trajectory." A rollout size of 2048 the agent will act for 2048 steps, collect all the states, actions, and rewards from that episode, and update its policy based on that large chunk of data later. This large buffer of data guarantees the updates are

done from a wide range of contexts, improving stability.

For the batch size, Once it has received the 2048 steps of experience, the PPO algorithm does not feed them all into the neural network simultaneously. They are split into mini batches of smaller quantity. The batch size is the quantity of mini batches. The approach, which is referred to as mini batch gradient descent, is much more computationally intensive and stable compared to trying to update all 2048 samples at once.

The training set consisted of 70 percent of the total available amount of stock data, with the other 30 percent split between validation (15 percent) and test (15 percent). The agent learned on over 50,000 timesteps, where it gained the capability to optimize rewards from trading actions based on previously observed price trends and indicators. The policy network (MlpPolicy) employed a multilayer perceptron to map observations to probability distributions over actions. The clipped surrogate objective of the PPO allowed for stable updates without sharp policy parameter jumps at optimization.

4.3.1 Structure of the Neural Network

The Agent used the multilayer perception (MLP) policy, that comes with the stable baseline3 Library while training the PPO agent, recall that there is a rollout size of 2048 steps, which are split into mini batches, with a batch size of 64, before feeding them to the neural network, from this we have:

- Policy network (actor): which consists of 2 hidden layers, each with 64 units.

- Value network (critic): Seperate network but same structure, which consists of 2 hidden layers each with 64 units

The structure looks like

Input (state features) $\rightarrow [64] \rightarrow [64] \rightarrow$ Output layer (actions / state value)

So the Actor network outputs a probability distribution over the 3 actions (Buy, Sell, Hold), and the critic network outputs a scalar value estimate of the state (expected return), then finally the PPO then optimizes both together with its clipped objective.

4.4 Evaluation

The trained agent was tested in the unseen testing environment, with the following performance metrics:

- Cumulative Return (CR):

  $CR = \frac{V_{final} - V_{initial}}{V_{initial}}$

  where V_final is the final portfolio value and $V_initial$ is the initial or starting balance

- Sharpe Ratio (SR):

  $SR = \frac{\mu_r}{\sigma_r}\sqrt{252}$

  where μ_r is the average of daily returns,σ_r is the standard deviation of daily returns, and 252 is the number of trading days in a year. This captures risk adjusted return, with a factor of excess return earned over volatility.

These metrics provide both absolute and risk adjusted views of trading performance.

4.5 Results

In testing, the following performance was attained by the agent:

- Cumulative Return: 1.4163
- Sharpe Ratio: 0.1648



Figure 4.1: Results

The cumulative return suggests that the PPO based agent was able to make profits from the initial portfolio many times during the test period. However, the relatively low Sharpe ratio (close to zero) indicates that the returns generated were achieved with very high volatility, i.e., the strategy was not quite efficient at risk adjusted return generation.

The net worth profile showed that the agent was able to capture profitable opportunities, but it also experienced oscillations(fluctuations) in accordance with the inherent volatility of the Nigerian stock market. This implies that while the agent's trading strategy was profitable in an absolute sense, it was not firmly consistent relative to the risks taken.

Financially, a Sharpe ratio closer to or above 1 is preferable, suggesting more stable excess returns per unit of risk. Therefore, while the results are suggestive of the potential for PPO to uncover profitable patterns, they also suggest the potential for further hyperparameter tuning, longer training horizons, or the inclusion of risk sensitive reward functions to improve stability and risk adjusted performance. The strategy learnt by this agent to yeild this cummulative return, might not be replicated when the agent is trained again, based on the sharpe ratio gotten the agent is still taking risks and the agent was just lucky at this point to yeild such increase, at best to avoid a loss the agent might just stay without trading since it is not sure of the strategy it has learnt, that would then yeild a cummulative return of 0.0000. To improve this, we then have to do certain things like hyperparameter tuning, adding macro economic data like inflation rate etc and even calculating more technical indicators to make the agent gain more information, also reshaping the reward function as well as action space which currently is just go in all in buying and selling. creating a different action space that is continuous (e.g, buying shares with 30 percent of balance at specific times) instead of just discrete might encourage the agent trade more carefully since its not going all in.

This strategy by the PPO agent was compared with a buy and hold strategy for a period of one year also, investing the same amount of 10000 naira, the amount was split across 50 selected companies and 200 naira was invested, after the period of one year the 10000 grew to 11,480.07 naira, which means a profit but with a cummulative return of 0.1480. The PPO agent learnt a strategy that was more profitable but was more exposed to risks and the volatility of the Nigerian stock market.

The chart in figure 4.1 shows the fluctuation in net worth, how the trained agent starts of by making progress, makes some losses and continuously in that manner.

The chart in 4.2 indicates the performance of the 50 selected company in terms of profit or loss made on the original investment. The bars are green for profit and red for loss, and it is immediately clear which investments were profitable. ACADEMY

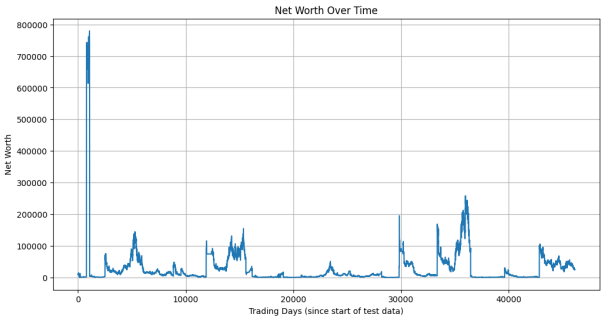


Figure 4.2: Networth over Time

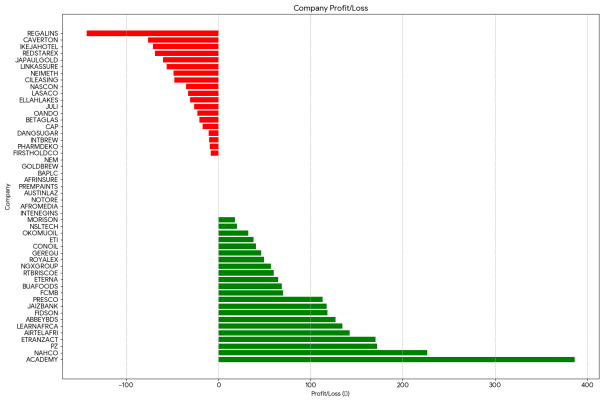


Figure 4.3: Visuals of 50 companies profit/losses

recorded the highest profit, followed by NAHCO and ETRANZACT. REGALINS recorded the maximum loss, followed by CAVERTON and IKEJAHOTEL. AUSTINLAZ, NOTORE, and INTENEGINS were some of the companies whose profit/loss was zero, i.e., their last value equaled the initial investment.

CHAPTER 5

Conclusions and Recommendation

5.1 Conclusion

The thesis sought to design and experiment with an intelligent Nigerian Stock Market trading agent using Deep Reinforcement Learning (DRL), with the Proximal Policy Optimization (PPO) algorithm as the methodology's cornerstone. By developing a customized trading environment in Gymnasium, training the PPO agent with historical NGX stock data (2000-2025), and benchmarking with a buy and hold strategy, the research has demonstrated both the potential as well as the limitations of DRL in emerging market contexts such as Nigeria.

The result indicated that the PPO trading agent had a total return of 1.4163, meaning an initial investment of ₦10,000 grew to approximately ₦24,163, a profit of ₦14,163. In contrast, a buy-and-hold strategy applied to 50 companies in 2022 gave a portfolio value of ₦11,480.07, equivalent to a profit of ₦1,480.07. This indicates that, at least from historical data, the PPO agent performed much better than the traditional passive strategy.

However, even with this strong comeback, the Sharpe ratio of 0.1648 highlights an important weakness: the trading strategy was correlated with excess volatility over its returns. That is, the agent made money, but did so with a great deal of risk and volatility, and therefore the strategy is less trustworthy for real world application in this current form. This risk return trade off points to the broader problem in reinforcement learning

for financial markets: consistent, generalizable performance rather than fitting across the history of the data. The work thus contributes to the body of literature that DRL based agents can deliver greater returns in theory but must be heavily optimized for effective deployment in high risk, dynamic settings like stock trading.

5.2 Recommendation

Although promising, there are a few areas to enhance the performance, stability, and usability of the model for real world trading conditions. These suggestions range from algorithmic optimizations, feature engineering, and risk mitigation to deployment factors.

5.2.1 Enhanced Risk-Adjusted Performance

- Hyperparameter Tuning: The PPO agent was trained mostly using default hyperparameters in this study. Fine tuning parameters such as the discount factor, learning rate, clip ratio, and batch size may seriously stabilize training and reduce volatility in returns. Automatic hyperparameter tuning with methods such as Bayesian Optimization or Optuna based tuning can be applied to reveal optimal values.

- Large Training Timesteps: Scaling up the total training timesteps would allow the PPO agent to test more state action pairs, making it more general and restricting overfitting to short term market fluctuations.

- Reward Shaping: Modifying the reward function to punish high volatility or drawdowns would encourage the agent to aim for stable growth instead of short term profit.

5.2.2 Enhancing The Input Features

Introducing Additional Technical Indicators Besides SMA, RSI, and MACD, would make a difference by making the learned dataset more rich, additional advanced indicators such as Bollinger Bands, Average True Range (ATR), Stochastic Oscillator, and On-Balance Volume (OBV) could potentially provide more insightful signals for the agent,

also Including Macroeconomic Variables, would make the agent learn the effect of major factors that influences stocks in a market like Nigeria, Since the Nigerian stock market is driven by macroeconomic variables (e.g., interest rates, inflation, oil prices, exchange rates, and policy changes), including such variables in the state space could help the agent learn context sensitive policies.

The Nigerian system is a system with constant changing policies and day to day news on the economy, Sentiment Analysis which Includes news sentiment features, Twitter posts, or analyst recommendations could help to capture market psychology, which plays a crucial role in Nigerian equity trading.

5.2.3 Reducing Overfitting and Improving Generalization

Training the agent on not only NGX data but also on other African or emerging market exchanges can improve generalization, in that the agent won't overfit Nigerian specific trends, also Implementing dropout layers on the policy network or incorporating entropy regularization can nudge the agent to keep exploring and not converge too quickly to sub-optimal strategies.

5.2.4 Increasing Real World Rellevance

- Transaction Costs and Commisions: The environment currently has costless execution. In the real world, brokerage commission, liquidity etc., can erode profitability. Adding them to the environment would make the output more realistic.
- Risk Controls: Adding portfolio level risk constraints such as maximum drawdown limits, stop loss rules, or position size rules could make the strategy more practically resilient.
- Ensemble Models: Mixing PPO with some other DRL algorithms such as Deep Q-Networks (DQN), Advantage Actor-Critic (A2C), or Soft Actor-Critic (SAC) can form a hybrid agent that offers exploration, exploitation, and risk management.

5.2.5 Broader Proposals for Future Work

- Live Market Testing: The ultimate test of such an agent is live deployment with live NGX data. Future work should include live trading deployment in a paper

trading environment initially before going to live trading.

- Scalability Studies: Experimentation on how the model performs on larger capital bases and in multi asset classes (equities, bonds, derivatives) could make it more applicable.
- Explainability and Interpretability: One of the largest hurdles to financial practical adoption for DRL is the "black box" issue with models. Developing explainable RL (XRL) methods for determining why the agent takes a specific action would be important for investor trust and regulatory needs.
- Integration with Portfolio Management Systems: Instead of trading individual stocks, scaling the agent up to have portfolio level optimization might put it more into institutional use.

5.3 Final Remarks

The study has indicated that DRL, specifically PPO, is capable of delivering improved returns in the Nigerian stock market compared to traditional approaches. Profitability, however, is not sufficient; risk adjusted performance must be improved prior to such systems being poised to transition from experimental frameworks into production level instruments for financial institutions, investors, and traders, as even the results gotten now is not sure to be replicated next time as the agent gained based on luck and not completely through a learned startegy.

By optimizing hyperparameters, enriching feature inputs with technical and macroeconomic data, incorporating realistic frictions in the market, and injecting strong risk controls, future implementations of this research could deliver trading agents capable of not only beating the market, but doing so safely, reliably, and transparently.

Lastly, the dream is to bridge scholarly research on reinforcement learning and practical financial innovation in the emerging markets with a view of paving the way for wiser, more flexible, and more robust trading systems.

APPENDIX A

Technical Indicators

Simple Moving Average (SMA)

20-day SMA:

$$\text{SMA}_{20} = \frac{P_1 + P_2 + \cdots + P_{20}}{20}$$

50-day SMA:

$$\text{SMA}_{50} = \frac{P_1 + P_2 + \cdots + P_{50}}{50}$$

Relative Strength Index (RSI)

Step 1: Price change (ΔP)

$$\Delta P_t = P_t - P_{t-1}$$

Step 2: Separate gains and losses

$$\text{Gain}_t = \begin{cases} \Delta P_t & \text{if } \Delta P_t > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{Loss}_t = \begin{cases} -\Delta P_t & \text{if } \Delta P_t < 0 \\ 0 & \text{otherwise} \end{cases}$$

Step 3: Average gain and loss (using exponential moving average, EMA)

$$\text{Avg Gain}_t = \text{EMA}_{14}(\text{Gain})$$

$$\text{Avg Loss}_t = \text{EMA}_{14}(\text{Loss})$$

Step 4: Relative Strength (RS)

$$RS_t = \frac{\text{Avg Gain}_t}{\text{Avg Loss}_t}$$

Step 5: RSI

$$RSI_t = 100 - \frac{100}{1 + RS_t}$$

Moving Average Convergence Divergence (MACD)

Step 1: Calculate EMAs

$$EMA_{12} = \text{EMA with span 12 of price}$$

$$EMA_{26} = \text{EMA with span 26 of price}$$

Step 2: MACD Line

$$MACD = EMA_{12} - EMA_{26}$$

Step 3: Signal Line

$$\text{MACD_Signal} = EMA_9(MACD)$$

Where the 9-period EMA is applied to the MACD line to create the signal line.

APPENDIX B

Another Appendix

$$\text{Cumulative Return} = \frac{\text{Final Net Worth} - \text{Initial Balance}}{\text{Initial Balance}}$$

$$\text{Sharpe Ratio} = \frac{\overline{R_d}}{\sigma_d} \times \sqrt{252}$$

The data set was gotten from Kaggle titled the Nigerian Financial Market Data (2000-2025)

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